

Distributed Quantum Error Correction: Advancements and Future Research Directions

1st Shahram Babaie, 2nd Sean Grzenda, 3rd Chunming Qiao

Department of Computer Science and Engineering

University at Buffalo, Buffalo, USA

{shahramb, seangrze, qiao}@buffalo.edu

Abstract—The transition from monolithic quantum computing architectures to distributed quantum computing (DQC) is inevitable due to the limited number of qubits available in a single quantum processing unit (QPU) and the growing demand for utility-scale, fault-tolerant quantum systems. As a part of this transition, distributed quantum error correction (DQEC) emerges as a foundational component for addressing the error-prone nature of qubit storage and gate operations across multiple QPUs. In this paper, we provide a comprehensive review of recent advancements in the DQEC code families, including both topological and hypergraph product codes. We highlight how DQEC frameworks can enhance effective computational capacity and enable the realization of complex and scalable quantum codes on distributed quantum platforms. We also discuss the advantages and constraints associated with various stages of DQEC, including code design, encoding, syndrome extraction, and decoding. Furthermore, we outline key technical challenges and explore promising directions for future research to emphasize the pivotal role of DQC in advancing DQEC codes and shaping the development of future-generation quantum error correction codes.

Index Terms—Quantum error correction, Distributed quantum computing, Monolithic quantum computing, Hypergraph product code, topological codes

I. INTRODUCTION

Quantum computing harnesses quantum mechanics to tackle problems that are intractable for state-of-the-art (SOTA) digital computing counterparts. However, realizing its potential entirely depends on addressing its error-prone nature and resource constraints. Current systems, known as Noisy Intermediate-Scale Quantum (NISQ) devices, are vulnerable to errors arising from faulty state preparation, probabilistic measurements, defective quantum gates, imperfect qubits, and decoherence. Quantum Error Correction (QEC) plays an indispensable role in bridging the gap between physical error rates ($\approx 10^{-3}$) and the error rates ($\approx 10^{-15}$) required for large-scale quantum algorithms. Efficient *QEC codes* (QECCs), however, remain a central open challenge in the realization of practical, fault-tolerant quantum computing [1].

There is a growing consensus in both academia and industry that interconnecting multiple small-to-moderate-scale QPUs via a quantum data network (QDN) offers a viable path toward overcoming the scalability limitations of *monolithic quantum computing* (MQC). By distributing quantum computation across interconnected QPUs, DQC enables the aggregation of quantum resources and offers improved modularity and scalability. However, it faces implementation hurdles, including

the need for high-fidelity inter-QPU communications, reliable entanglement distribution, and system-level optimization tasks involving circuit partitioning, qubit allocation, and load balancing. Moreover, the distributed nature of such architectures reshapes the error landscape, as errors arise not only from local quantum operations but also from communication and entanglement routing. As a result, DQC naturally raises the critical question of how QEC can be designed, implemented, and optimized for distributed environments [2], which motivates the study of *distributed QEC* (DQEC) and the development of DQEC codes (DQECCs).

The shift toward DQEC is also driven by the technical challenges associated with long-range gates [3] and multi-planar architectures [4], which are required for the efficient implementation of quantum low-density parity-check (qLDPC) and hyperbolic quantum codes. Conventional qubit-based QEC schemes rely on encoding a single *logical qubit* (LQ) into multiple physical qubits to construct fault-tolerant qubits, which drives up the demand for physical qubits. Moreover, error correction procedures require extra ancilla qubits and redundant quantum gates for encoding and syndrome extraction, which introduce new sources of error, necessitating further qubit and gate overhead to surpass their impact. DQECCs offer a promising avenue to address these challenges by enabling access to expanded quantum resources, such as quantum memory, non-local quantum gates, and parallelism. However, realizing these benefits requires effective design across all stages of the error correction process, including code construction, syndrome extraction, decoding, error modeling, and optimization [5].

In this paper, we explore the parallel evolution of DQECCs alongside the transition toward DQC. We not only review recent advancements in QECCs, but also highlight the critical role of DQEC in streamlining code design, decoding, and syndrome extraction for both topological codes and hypergraph product (HGP) codes. In particular, given the structural differences among qubit-based QECCs, DQEC influences each code family in distinct ways. Accordingly, we first briefly review recent works on algebraic codes, including the three-qubit code and Calderbank–Shor–Steane (CSS) codes, and then we focus on prominent topological codes, such as toric, color, surface, and Floquet codes, as well as several classes of HGP codes, such as hypergraph product, fiber bundle, and balanced product codes. Finally, we delineate the outstanding

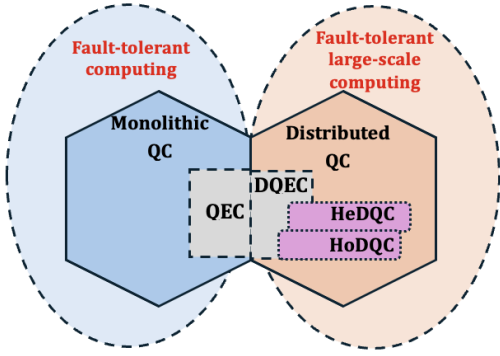


Fig. 1: A general overview of survey scope.

challenges associated with each code family and outline a roadmap toward scalable and fault-tolerant DQEC codes. Figure 1 illustrates the scope of this study that details the shift from monolithic quantum architectures toward DQC, which includes both *heterogeneous DQC* (HeDQC) and *homogeneous DQC* (HoDQC). It highlights the central role of QEC and DQEC in the achievement of fault-tolerant computing systems.

The remainder of this paper is organized as follows. Section II describes fundamental concepts associated with error models and stabilizer formalism. Sections III and IV explore distributed topological codes and distributed HGP codes, respectively. Section V outlines open challenges and prospective research directions, followed by the conclusion in Section VI.

II. ERROR MODELS AND PRELIMINARIES

A. Error models and Logical qubits

Quantum computations are susceptible to errors arising from imperfect state preparation, gate operations, and measurement, which degrade the fidelity of quantum states and operations. Faulty state preparation leaves the system in a mixed state, $\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$, instead of the intended pure state. Probabilistic measurements also unavoidably introduce readout errors, which necessitate repeated measurements to achieve reliable outcomes. Moreover, qubits are non-isolated and tend to interact with the environment, which results in decoherence and loss of the coherent superposition state, and can be modeled as Equation 1.

$$\begin{aligned} |0\rangle_Q |0\rangle_E &\rightarrow |0\rangle_Q |0\rangle_E \\ |1\rangle_Q |0\rangle_E &\rightarrow \sqrt{1-\omega} |1\rangle_Q |0\rangle_E + \sqrt{\omega} |0\rangle_Q |1\rangle_E \end{aligned} \quad (1)$$

where $\omega = 1 - e^{-\frac{t}{T_1}}$ denotes the damping probability and T_1 is the relaxation time. Although imperfect quantum gates can induce small under-rotation or over-rotation, the coherent errors can accumulate quadratically with circuit depth and distort the expected output, necessitating the need for robust QEC and fault-tolerant designs.

The quantum noise processes are classified into coherent and incoherent errors. Coherent errors, such as faulty gates, arise from systematic and deterministic imperfections in control operations, whereas incoherent errors, such as State preparation and measurement (SPAM), result from environmental

interactions. Incoherent errors are typically modeled using the Pauli channel, where Pauli operators ($\sigma_x, \sigma_y, \sigma_z$) represent different error types. The Pauli X-error, referred to as a bit-flip error, swaps the computational basis states, the Pauli Z-error, or phase-flip error, exchanges the Hadamard basis eigenstates, and the Pauli Y-error corresponds to the simultaneous action of X- and Z-errors up to a global phase.

In practice, the implementation of reliable physical qubits is fundamentally constrained by incomplete quantum theories, imperfect hardware components, and their tendency to interact with the surrounding environment. A near-term solution to mitigate these effects is to encode quantum information into logical qubits, formed by jointly entangling multiple physical qubits through QECCs. For instance, Steane and Shor codes require seven and nine physical qubits per logical qubit, whereas surface codes may require on the order of thousands of physical qubits per logical qubit [6]. In MQC architectures, all logical qubits and their associated syndrome extraction (SE) circuits reside on a single QPU, as illustrated in Fig. 2 a. Although this design benefits from local operations and simplified control, it is constrained by the finite number of physical qubits and intra-QPU resources, which bound the number of logical qubits.

Distributed quantum computing addresses these limitations by allowing multiple QPUs to cooperatively execute quantum workloads and share physical qubits. As shown in Fig. 2 b, each QPU hosts a subset of physical qubits and a local syndrome extraction circuit. Logical qubits in this architecture may be constructed either entirely within a single QPU (e.g., LQ#1 through LQ#i) or in a distributed manner using physical qubits residing on multiple QPUs (e.g., LQ#j+2). The coordination among QPUs is achieved through a control backbone composed of three primary components, i.e., classical coordination electronics, syndrome analysis, and a shared cryogenic backplane. The cryogenic backplane provides an ultra-low-temperature environment for hosting multiple QPUs and handles controls, readout signals, and entanglement distribution, while maintaining coherence. The classical coordination electronics act as a classical control hub to connect QPUs to classical readout systems. Syndrome analysis aggregates measurement outcomes of local SE circuits, performs decoding, and performs intra-QPU syndrome extraction [7].

B. Stabilizer formalism and CSS codes

A quantum code is typically characterized by parameters $[[n, k, d]]$, where n is the number of physical qubits, k is the number of encoded (logical) qubits, and d is the code distance, defined as the minimum weight of a Pauli operator (i.e., referred to as logical operator) that acts nontrivially on the codespace while remaining undetectable by the stabilizer measurements. In general, quantum error correction codes utilize entangled cat-like states and stabilizer measurements to address the challenges posed by the no-cloning theorem and measurement-induced state collapse, respectively. Babaie and Qiao [8] proposed a distributed implementation of an entangled cat-like state across three QPUs, in which each

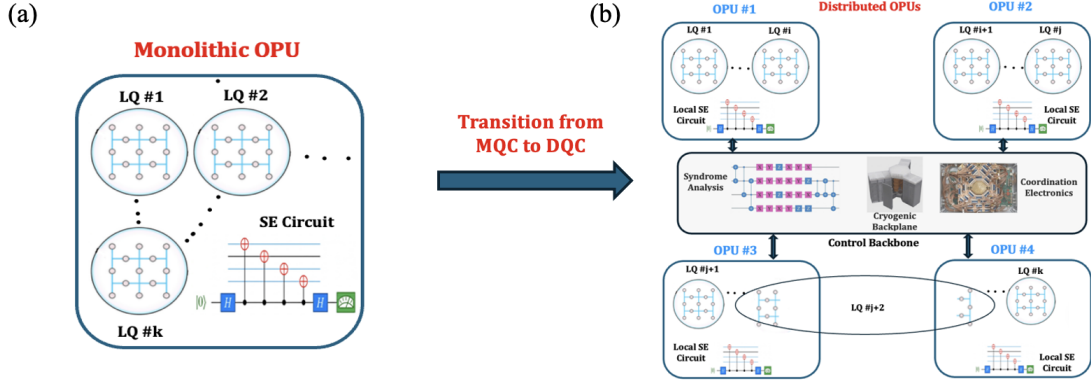


Fig. 2: Quantum error correction concept in MQC and DQC.

QPU contributes a subset of its qubits to construct a fault-tolerant logical qubit. This architecture allows partially faulty QPUs, which are referred to as QPUs containing a mixture of faulty and functional qubits, to participate in the construction of logical qubits. The authors validated the correctness of the distributed joint quantum state for both bit-flip and phase-flip error correction using the Qiskit framework. Furthermore, they assessed the effectiveness of distributed logical encoded states in terms of fidelity, hardware complexity, and trade-offs between error probability and computational accuracy.

Stabilizer codes provide a unifying framework for describing a broad class of QECCs. In this formalism, stabilizers are mutually commuting Pauli operators that leave the codespace invariant, and their measurement outcomes reveal the presence and nature of errors. The set of such operators forms an Abelian stabilizer group (excluding global phases), $\mathcal{S} = \langle S_1, S_2, \dots, S_r \rangle \subset \mathcal{P}_N \setminus \{\pm i, -1\}$, where $\mathcal{P}_N$ denotes the N -qubit Pauli group. A quantum state $|\psi\rangle$ belongs to the codespace $Q(\mathcal{S})$ if and only if it is a simultaneous $+1$ eigenstate of all stabilizer generators (i.e., $S_i|\psi\rangle = |\psi\rangle \forall i$). Logical operators are elements of the normalizer of $\mathcal{S}$ that commute with all stabilizers but are not members of $\mathcal{S}$, and act nontrivially on the codespace.

The syndrome of a Pauli error $E \in \mathcal{P}_N$ is a binary vector $\sigma(E) = (\sigma_1, \dots, \sigma_r) \in \{0, 1\}^r$, where σ_i indicates whether the error commutes or anticommutes with the corresponding stabilizer generator S_i as Equation 2.

$$\sigma_i = \begin{cases} 0 & \text{if } ES_i = S_iE \text{ (commute)} \\ 1 & \text{if } ES_i = -S_iE \text{ (anticommute)} \end{cases} \quad (2)$$

where r denotes the number of independent stabilizer generators. A notable subclass of stabilizer codes is the CSS codes, in which the stabilizer group contains only X-type and Z-type operators. These are defined by two binary matrices H_X and H_Z , which are responsible for phase-flip and bit-flip detection, respectively. These matrices must satisfy the orthogonality constraint $H_X H_Z^T = 0 \pmod 2$, which ensures the commutativity of the associated stabilizers. A CSS code can be constructed from two classical linear codes, $C_X = \ker(H_X)$ and $C_Z = \ker(H_Z)$, such that $C_Z^\perp \subseteq C_X$, and is denoted

by $\text{CSS}(C_X, C_Z)$. A CSS code has parameters $[[n, k, d]]$, where $k = n - \text{rank}(H_X) - \text{rank}(H_Z)$ is the number of encoded qubits. The distance is defined as $d = \min(d_X, d_Z)$, where $d_X := \min_{c \in C_X \setminus C_Z^\perp} \|c\|$ and $d_Z := \min_{c \in C_Z \setminus C_X^\perp} \|c\|$, corresponding the minimum weight of vectors in the set of $(C_X \setminus C_Z^\perp) \cup (C_Z \setminus C_X^\perp)$. Moreover, it has been shown that any $[[n, k, d]]$ stabilizer code can be mapped to a $[[4n, 2k, 2d]]$ CSS code [9].

The distributed implementation of CSS code, as a fundamental component of many topological quantum codes, has been proposed by Babaie and Qiao [10]. In this architecture, Steane's code, which includes six stabilizer generators, is distributed across seven QPUs using a distributed Greenberger-Horne-Zeilinger (GHZ) state. The original quantum state is encoded as superpositions of even-weight and odd-weight components of the CSS code. The authors demonstrated the effectiveness of their method in detecting X-errors, Z-errors, and Y-errors, which highlights the robustness of the distributed proposed scheme.

III. DISTRIBUTION OF TOPOLOGICAL CODES

Aligned with recent industrial advances in DQC, such as IBM Quantum System II, current research efforts have focused on minimizing distribution overhead, such as the number of entangled EPR (Einstein-Podolsky-Rosen) pairs, non-local operations, ancilla qubits, and encoding/decoding complexities. Although non-local operations, such as non-local CNOT (Figure 3 a) and non-local CZ (Figure 3 b), and quantum teleportation (Figure 3 c) are utilized in DQECCs, optimizing these mechanisms to address performance, resource efficiency, and fault tolerance remains an open research problem.

In general, topological codes constitute a prominent family of QECCs, in which quantum information is encoded into topological features of geometric manifolds, such as toroidal or planar surfaces. Stabilizer generators of these codes are geometrically local, and logical operators correspond to non-contractible topological loops, which are robust against small uncorrelated errors. The toric code, for example, extends the intuition of the classical repetition code via the Cartesian product of two cyclic codes, which independently correct

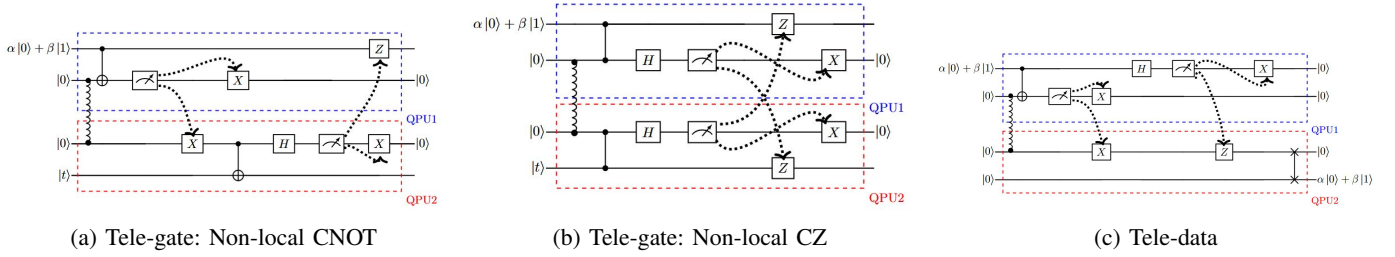


Fig. 3: Distributed quantum computing components.

bit-flip and phase-flip errors. The number of logical qubits is determined by the number of independent homologically nontrivial cycles, and the code distance is determined by the length of the shortest non-contractible loops and scales linearly with the lattice dimension as $d = \Theta(\sqrt{2L^2}) = \Theta(L)$, where $2L^2$ is the number of physical qubits on $L \times L$ square lattice torus. The surface code, a planar variant of the toric code, replaces periodic boundaries with open boundaries, which makes it more practical for implementation while preserving favorable code parameters. Topological codes are considered the leading candidate for realizing utility-scale, fault-tolerant quantum computing architectures due to the local stabilizer structure and favorable error thresholds [6].

Experimental demonstrations have validated the practicality of surface code across multiple hardware platforms. Google Quantum AI [11] reported experimental results on the Willow chip showing exponential suppression of logical errors as code distance increased. Their experiments demonstrated that increasing the code distance by two reduces the logical error rate by more than a factor of two, and a distance-seven code achieves a logical error rate of 0.143%. These results represent a significant milestone toward scalable, fault-tolerant quantum computing based on surface code architectures.

The success of quantum error correction depends not only on code construction but also on the performance of the associated decoding algorithm. Decoding identifies the appropriate recovery operation (R) by processing error-sensitive events (ESEs), which arise from data-qubit errors and measurement errors, and are detected through changes in the results of the stabilizer generators. These events are mapped onto a decoding hypergraph, where the vertices correspond to detection events, and the hyperedges encode correlated error mechanisms imposed by the structure of the code and the syndrome extraction circuits. Depending on the topology and structure of the hypergraph, various decoding algorithms, such as matching-based, belief-propagation, and tensor-network decoders, can be leveraged to infer the most likely recovery operations. The decoding of CSS codes can be decomposed into two independent subsections, i.e., X-decoder and Z-decoder. Let E denote the actual physical error and R_X and R_Z the corresponding recovery operations. The residual error is given by $\Gamma = R_X R_Z E$. Decoding algorithm is successful if Γ acts trivially on the codespace (i.e., $\Gamma \in \langle S \rangle$) and the logical state is the $+1$ eigenstate of the residual error (i.e.,

$\Gamma|\psi\rangle_L = (+1)|\psi\rangle_L$). In contrast, decoding fails if Γ is logically nontrivial (i.e., $\Gamma \in \langle S \rangle L \setminus \langle S \rangle$) and results in a logical error on the encoded qubit.

Decoding algorithms such as minimum-weight perfect matching (MWPM) and cluster-growth techniques (e.g., union-find) exploit structured correlations among ESEs to infer the most likely error chains. Li et al. [12] proposed a generative adversarial network-based decoder that achieves a higher fidelity threshold ($P=0.2108$) than classical decoding methods ($P=0.1099$). Ji et al. [13] demonstrated that graph attention networks can improve accuracy by approximately 13% for small lattice sizes. Ao-Qing Li et al. [14] further improved decoding performance using transformer-based models. Although neural-network-based decoders offer performance advantages, they often incur higher computational latency, which can limit their applicability in real-time scenarios. Wang et al. [15] addressed this challenge by designing a parallel feed-forward neural network architecture that reduces inference latency while still outperforming MWPM in terms of decoding accuracy. Although these methods were developed for surface code, many of them can be adapted to broader classes of topological codes.

High-fidelity single- and two-qubit gate operations are essential for practical deployments of topological codes. Lattice braiding is an effective solution for optimizing logical gate operations, in which logical qubits are moved along predefined trajectories that can be abstracted as braiding paths or virtual pipes. However, a fundamental constraint of this approach is that simultaneous braiding paths cannot cross, which complicates the problem of finding optimal braids. Wasa et al. [16] demonstrated that optimizing braiding paths is NP-Hard and can be reduced in polynomial time to the Planar-Rectilinear-3SAT problem. Hua et al. [17] proposed a dynamic logical-qubit allocation strategy that adjusts qubit placement during execution to reduce communication overhead. They claimed that this approach achieved up to a 30X speedup compared to conventional greedy strategies. An alternative paradigm for implementing logical operations is lattice surgery, which performs gates by merging and subsequently splitting logical patches, which typically incurs lower space-time overhead and offers improved layout flexibility compared to lattice braiding. Tan et al. [18] focused on these advantages in conjunction with SAT-based optimization to design optimized layouts and gate schedules for logical qubits.

Color codes constitute another class of two-dimensional topological stabilizer codes defined on a three-colorable lattice embedded in geometric surfaces. Although the lattice geometry determines the physical layout, the number of logical qubits and the code distance depend on the surface topology. The Euler characteristic of a closed and orientable lattice surface is given by $\chi = v - e + f$, where f , v , and e represent the number of plaquettes, vertices, and edges, respectively. Consequently, the number of encoded qubits is given by $k = 4 - 2\chi = 2g$, where g is the genus of the lattice. The code distance corresponds to the shortest nontrivial homology cycle on the surface, which is equivalent to the minimum weight of a nontrivial logical operator. Chandra et al. [19] proposed a distributed implementation of the (6,6,6) color code, known as the hexagonal code, across multiple QPUs. In this architecture, stabilizer measurements, associated with boundary checks, are implemented using non-local CNOT gates and evaluated under an asymmetric noise model. Simulation results for code distances up to 15 indicated that probability threshold values of approximately 0.10533% for matrix product state (MPS) decoding and 0.09809% for concatenated MWPM decoding, which are significantly lower than those of monolithic implementations. However, the exploration of alternative lattice geometries for distributed color codes, the development of decoders optimized for asymmetric noise, and the improvement of the accuracy of quantum network simulation models remain open research issues.

Bone et al. [20] introduced a fully distributed implementation of the toric code using GHZ states to construct distributed stabilizer measurements. In this architecture, dynamic programming is utilized to construct large GHZ states from smaller ones through fusion or entanglement distillation operations. Compared to the monolithic toric code, the distributed implementation demonstrates significantly lower error thresholds. Despite these promising results, formalizing the binary tree into an explicit protocol and developing a dynamic scheduler for the Bell pair distribution remain future research directions.

Floquet codes represent a class of QECCs in which logical qubits emerge dynamically through periodic sequences of stabilizer measurements [21]. In particular, honeycomb Floquet codes, which are implemented on a hexagonal lattice, and stabilizer generators are measured in a three-step periodic cycle (e.g., a six-round periodic cycle in X-Cube Floquet code) to enable error detection across both spatial and temporal dimensions. The number of logical qubits depends on the underlying topology, while code distance is determined by the minimum non-trivial cycle length. The unique features of Floquet codes, such as low-weight two-qubit measurements, dynamic on-demand construction of logical qubits, and cyclic stabilizer measurements, make them well-suited for new-term quantum devices. Kesselring et al. introduced the Floquet color code using anyon condensation [22], and both the honeycomb Floquet codes and the Floquet color codes have been experimentally tested on IBM quantum hardware. Further experimental results indicate that Floquet codes can

outperform surface codes on tailored architectures.

Sutcliffe et al. introduced a distributed implementation of the semi-hyperbolic Floquet code that leverages hyperbolic surfaces to achieve high encoding rates. They utilized fine-graining techniques, in which lattice edges are replaced by planar patches, to increase code distance from $O(\log(n))$ to $O(\sqrt{n})$, while preserving its topological properties. The resulting construction can be partitioned into different patches, where each patch can be assigned to separate QPUs. The authors claimed that these codes can tolerate realistic levels of both local and network noise (approximately 99.97% for local and 99% for non-local operations), which makes them a promising code family for scalable distributed quantum error correction. [23]. Alam and Rieffel introduced Floquet-Bacon-Shor codes, which integrate the low-overhead subsystem structure of Bacon-Shor codes with the cyclic measurement strategy of Floquet codes. The authors demonstrated that, on an $n = d \times d$ lattice with k dynamical logical qubits, the code distance scales as $\Theta(d/\sqrt{k})$, and these codes saturate the subsystem bound $kd = O(n)$ [24].

IV. DISTRIBUTION OF HGP CODES

Although topological codes such as the surface code are attractive for practical implementations due to their geometrically local stabilizers, they typically encode only a constant number of logical qubits and suffer from low encoding rates. In contrast, quantum LDPC (qLDPC) codes aim to overcome this limitation by constructing quantum codes from two classical linear codes (ξ_1 and ξ_2) through geometric constructions or hypergraph products. It has been shown that two asymptotically good classical codes (i.e., $\xi_1 = \xi_2 = [n, \Theta(n), \Theta(n)]$) can be combined to construct a good quantum code with parameters $Q = [[\Theta(n), \Theta(n), \Omega(\sqrt{n})]]$. This construction achieves linear scaling in both code length and the number of logical qubits, while maintaining a code distance that grows asymptotically with N .

Several constructions have been proposed to design qLDPC codes with improved code distance and code rate, which are summarized in Table I. Tillich and Zémor introduced the hypergraph product construction, which combines the Tanner graphs of two classical codes to achieve a quantum code with distance scaling as $d \propto \Theta(\sqrt{n})$ and a constant encoding rate. This construction requires orthogonal parity-check matrices H_X and H_Z satisfying $H_X H_Z^T = 0$. Let $T_X = (V_X, C_X)$ and $T_Z = (V_Z, C_Z)$ denote the Tanner graph representations of H_X and H_Z , respectively. The Cartesian product of these two Tanner graphs is given by $T_{XZ} = T_X \otimes T_Z$, with data qubits $V_{XZ} = (V_X \times V_Z) \cup (C_X \times C_Z)$, and stabilizers $C_{XZ} = (C_X \times V_Z) \cup (V_X \times C_Z)$ [25].

Evra et al. [26] broke the $\sqrt{n}$ distance barrier by incorporating the Kronecker-based product of good LDPC codes and high-dimensional expanders (HDX), in particular Ramanujan complexes. Their construction achieved $d \in \Omega(\sqrt{n} \log n)$ for three-dimensional Ramanujan complexes and $d \in \Omega(\sqrt{n} \log n)$ for two-dimensional Ramanujan complexes. The distance of qLDPC codes has been improved to

$d \in \Theta(\sqrt{n} \log^s n)$ using the iterated Kronecker-based product of two cosystolic expanders by Kaufman and Tessler [27]. Fiber bundle codes generalize tensor-product constructions by combining two classical codes in a base-fiber framework augmented with structured twists. This technique achieves a code distance scaling as $d \in \Theta(\text{polylog}(n)\sqrt{n})$, as shown by Hastings et al. [28]. The authors showed that random constructions create a family of qLDPC codes with $k \in \Theta(n^{3/5})$ and $d \in \Omega(n^{3/5}/\text{polylog}(n))$. Panteleev and Kalachev [29] developed lifted product codes to achieve almost linear scales, $k \in \Theta(n^\alpha \log(n))$ and $d \in \Omega(\frac{n^{1-\alpha/2}}{\log(n)})$, $0 \leq \alpha < 1$. Breuckmann and Eberhardt introduced balanced product codes using a cyclic symmetry action to scale the distance and dimension of qLDPC to $d \in \Omega(n^{3/5})$ and $k \in \Theta(n^{4/5})$, respectively [30].

Although qLDPC codes offer strong theoretical performance, effective decoding remains a challenge. Conventional belief propagation (BP)-based decoders are generally unsuitable for qLDPC codes because of trapping sets arising from short loops in the Tanner graph. Several techniques, such as message format modifications, stabilizer inactivation, optimized variable-node selection, and neural belief propagation, have been proposed to address this problem.

Implementation of qLDPC codes on superconducting hardware is challenging due to non-local stabilizers and hardware connectivity constraints. Tremblay et al. introduced an l -planar structure that partitions the Tanner graph into $\lceil \frac{\delta}{2} \rceil$ subgraphs, where δ is the maximum degree of the graph, to enable planar layouts without edge crossing. They also demonstrated that, by applying minimum edge coloring, each stabilizer can be measured with circuit depth $2\delta + 2$, while a reduced depth of $\delta + 2$ can be achieved with HGP codes [4]. In addition, long-range CZ gates implemented via resonator-induced phase gates offer a practical component for the implementation of qLDPC codes [3]. An alternative architecture based on atom arrays offers dynamic reconfiguration of qubit connectivity during circuit evaluation while maintaining low decoherence. Strikis and Berent [31] presented a modular implementation of HGP codes by tailoring the hardware connectivity to the code structure. In their approach, a hypergraph product of a repetition code and a two-dimensional surface code, which defines the intra-modular connectivity and inter-modular connectivity, respectively, forms a three-dimensional surface code. This framework could be generalized by permitting heterogeneous intra-modular connectivity across multiple modules. These generalizations have the potential to enable the implementation of more efficient and higher-performing quantum codes within modular architectures.

V. OPEN ISSUES AND FUTURE DIRECTIONS

Although significant progress has been made in the design and implementation of DQECs, several open challenges and future research directions must be carefully addressed to realize robust and utility-scale, fault-tolerant quantum computing. In particular, efficient coordination of multiple QPUs

introduces new opportunities and constraints that necessitate careful architectural and algorithmic consideration.

• Inter-QPU Communication Overhead

Distribution of a quantum algorithm across multiple QPUs introduces inter-QPU operations and communication overhead, due to the need for high-fidelity entanglement and non-local operations. Although techniques, such as non-local operations and tele-data, effectively enable the collaboration of QPUs, they impose resource and latency costs. Therefore, innovative and efficient methods, including AI-based optimization techniques, graph-theoretic approaches, and neural graph models, can be leveraged to mitigate these overheads.

The problem of minimizing inter-QPU communication can be modeled as a Min-Cut or graph partitioning problem and can be optimized through classical algorithms such as iterative Ford-Fulkerson and Edmonds-Karp. In addition, global Min-Cut algorithms, including deterministic Stoer-Wagner and randomized Karger-Stein, provide robust tools for identifying communication bottlenecks and optimizing qubit placement. Moreover, machine learning-based heuristics, including neural graph models and reinforcement learning, and hardware-aware scheduling, offer effective strategies to optimize qubit placement and routing strategies under hardware constraints. These approaches can represent a promising direction toward scalable DQECs by strategically partitioning quantum operations and enhancing computational efficiency and performance.

• Inter-QPU Distance Constraints

Although entanglement-based communications offer a powerful mechanism for enabling DQC, their effectiveness is constrained by the fidelity of constructed entangled states. In particular, photon loss, decoherence, and fidelity of photon-based inter-QPU communications increase with communication distance, which degrades the entanglement quality and consequently the fidelity of distributed logical qubits. Moreover, photon loss grows exponentially as the transmission channel increases as a result of environmental interactions. These effects pose significant challenges for maintaining high-fidelity quantum correlations and reliable long-range distributed quantum systems.

Fast entanglement purification and distillation can be utilized to purify the entangled states. Moreover, graph-based scheduling algorithms, such as Floyd-Warshall and Johnson's algorithm, can optimize qubit placement by modeling inter-QPU connectivity. Furthermore, hierarchical and modular architectures, which cluster closely highly correlated QPUs or strongly interacting QPUs, can reduce distance-induced concerns and enable efficient implementation of robust and scalable DQEC codes.

• Hardware Heterogeneity

The heterogeneous nature of quantum hardware platforms presents both opportunities and challenges for the design and implementation of DQECs. Different qubit modalities, such as superconducting circuits, trapped ions, photonic systems, and quantum dots, exhibit distinct characteristics in terms of gate speed, coherence time, fidelity, measurement, and entanglement generation, which make certain modalities more

TABLE I: Comparative overview of QECCs for distributed quantum computing

Class	QECC	Distance	Dimension	DQEC Compatibility	Key Feature of Code
Algebraic	3-qubit Distributed Code	3	$O(n)$	Code construction	Fully distributed on different QPUs
	CSS Distributed Code	3	$O(n)$	Code construction	Based on distributed GHZ states
Topological	Toric Codes	$\Theta(\sqrt{n})$	2	Code construction	Cartesian product of two codes
	Surface Code	$\Theta(\sqrt{n})$	1	Decoding	Cartesian product of two codes with open boundaries
	Color Codes	$\Theta(\sqrt{n})$	1	Code construction	A three-colorable planar lattice using face-based stabilizers
	Hyperbolic Floquet Codes	$O(\sqrt{n})$	$O(n)$	Code construction	Based on fine-graining to increase distance, while preserving dimension
HGP	Hypergraph product Codes	$\Theta(\sqrt{n})$	$O(n)$	Code construction	Hypergraph product of two Tanner graphs
	Evra's Code	$\Omega(\sqrt{n} \log n)$	$O(n)$	Not yet explored	Based on high-dimensional expanders (Ramanujan complexes)
	Kaufman's Code	$\Theta(\sqrt{n} \log^s n)$	$O(n)$	Not yet explored	Iterated tensor products of two cosystolic expanders
	Fiber Bundle Codes	$\Omega\left(\frac{n^{3/5}}{\text{polylog}(n)}\right)$	$\Theta(n^{3/5})$	Not yet explored	Twisted product of a random classical code and a cycle graph
	Lifted Product Codes	$\Omega\left(\frac{n^{1-\alpha/2}}{\log(n)}\right)$	$\Theta(n^\alpha \log(n))$	Not yet explored	Structured lifting of classical Tanner graphs
	Balanced Product Codes	$\Omega(n^{3/5})$	$\Theta(n^{4/5})$	Not yet explored	Balanced product construction with cyclic symmetry

suitable for specific roles in DQEC codes. For instance, data qubits, which must preserve quantum information over long durations, require long coherence times and low error rates, as well as the ability to sustain multiple iterations of stabilizer measurements. These requirements are typically met by technologies such as trapped ions, neutral atoms, and superconducting qubits. In contrast, measurement (ancilla) qubits, which are utilized for X- and Z-type stabilizer extraction, require fast, high-fidelity operations, rapid readout, and efficient reset mechanisms. Consequently, superconducting qubits and spin qubits are suitable for these qubits.

Heterogeneous quantum architectures offer a promising pathway toward scalable and high-performance DQECCs by assigning qubit roles according to the strengths of each QPU. However, it introduces significant integration and system-level challenges. Addressing this heterogeneity requires rigorous standardization of communication interfaces, precise synchronization, and robust control protocols to ensure seamless interoperability and reliable distributed quantum systems.

• Hybrid Quantum-classical Systems

Although the distribution of QECCs across multiple QPUs mitigates the scalability limitations of MQC, it introduces new challenges associated with the integration and coordination of quantum and classical resources. In particular, decoding, syndrome processing, and feedback control rely on classical computation and high-performance computing (HPC), which introduces several system-level challenges related to conversion overhead, mismatched processing speeds, timing control, scalability, and throughput. These issues not only degrade accuracy, introduce latency, disrupt synchronization, but also impair fidelity and increase decoherence.

Classical controllers must handle complex coordination problems as the number of QPUs scales, which can limit scalability. Therefore, the communication and computational

complexities of the classical controllers must scale approximately linearly with the number of QPUs. These challenges are exacerbated in heterogeneous DQC architectures, where QPUs may differ in hardware characteristics. Several solutions, such as low-latency communication networks, parallel decoding strategies, latency-aware compilers, network-aware qubit mapping, and robust synchronization protocols, can be utilized to address the challenges associated with the implementation of DQEC codes in hybrid quantum-classical systems.

• Entanglement Routing

Entanglement generation and distribution constitute foundational components of DQECCs, as they enable non-local operations and tele-data across spatially separated qubits. Practical DQECCs require both pre-distributed and on-demand entanglement. Although larger multipartite entangled states can be constructed from smaller ones through techniques such as Fusion-Retain-Only (FRO) and General Fusion (GF), these processes must be carefully integrated with entanglement purification protocols to preserve high fidelity.

Graph-state representations offer a flexible framework for constructing entanglement across arbitrary qubits, where intermediate qubits play a vital role in entanglement propagation. In general, effective DQEC therefore depends on adaptive entanglement routing and resource management that account for decoherence, loss, and network variability, ensuring reliable and scalable quantum communication.

• Distributed Decoding

Decoding algorithms infer the most likely error configuration from the measured syndromes and determine the appropriate correction operations. In large-scale and distributed quantum systems, this task becomes increasingly challenging due to the growing code complexity and stringent timing constraints. Conventional decoding algorithms suffer from high computational overhead and exhibit limited scalability as

system size increases, and standard belief-propagation-based decoders perform poorly for qLDPC codes due to the presence of short cycles and trapping sets, which hinder convergence and degrade decoding accuracy.

Distributed decoding strategies and machine learning-based approaches have emerged as promising alternatives to address the mentioned limitations. Neural network-based and reinforcement learning decoders have demonstrated improved performance in handling complex error patterns and adapting to nontrivial noise models. However, designing scalable and robust decoders remains an open challenge, as they must be tailored to the underlying code structure, including topological codes and HGP codes.

VI. CONCLUSION

Distributed quantum error correction represents a promising paradigm for realizing scalable and fault-tolerant quantum computation by leveraging the resources of spatially separated QPUs. In this study, we have explored the foundational aspects of DQC architectures, including code construction, syndrome extraction, and decoding, to enable QECCs across topological and hypergraph product code families. Although modular and distributed architectures mitigate several limitations associated with MQC, they also introduce new challenges, including inter-QPU communication overhead, entanglement routing, quantum-classical coordination, and scalable decoding, which have been reviewed to date. Moreover, we have reviewed recent advances spanning theoretical foundations, architectural designs, and practical implementations as critical steps toward the realization of DQECCs. Furthermore, we have highlighted open issues that will shape the next generation of scalable, robust, and fault-tolerant quantum computing systems.

REFERENCES

- [1] E. Campbell, "A series of fast-paced advances in quantum error correction," *Nature Reviews Physics*, vol. 6, no. 3, pp. 160–161, 2024.
- [2] D. Barral, F. J. Cardama, G. Díaz-Camacho, D. Faílde, I. F. Llovo, M. Mussa-Juane, J. Vázquez-Pérez, J. Villasuso, C. Piñeiro, N. Costas, J. C. Pichel, T. F. Pena, and A. Gómez, "Review of distributed quantum computing: From single qpu to high performance quantum computing," *Computer Science Review*, vol. 57, p. 100747, 2025.
- [3] X. Deng, W. Zheng, X. Liao, H. Zhou, Y. Ge, J. Zhao, D. Lan, X. Tan, Y. Zhang, S. Li, and Y. Yu, "Long-range zz interaction via resonator-induced phase in superconducting qubits," *Physical Review Letters*, vol. 134, no. 2, 2025.
- [4] M. A. Tremblay, N. Delfosse, and M. E. Beverland, "Constant-overhead quantum error correction with thin planar connectivity," *Phys. Rev. Lett.*, vol. 129, p. 050504, 2022.
- [5] J. C. Boschero, N. M. P. Neumann, W. van der Schoot, T. Sijpesteijn, and R. Wezeman, "Distributed quantum computing: Applications and challenges," K. Arai, Ed. Cham: Springer Nature Switzerland, 2025, pp. 100–116.
- [6] A. G. Fowler, M. Mariantoni, J. M. Martinis, and A. N. Cleland, "Surface codes: Towards practical large-scale quantum computation," *Physical Review A*, vol. 86, no. 3, 2012.
- [7] S. Du, Y. Ding, and C. Qiao, "S-qgpu: Shared quantum gate processing unit for distributed quantum computing," *AVS Quantum Science*, vol. 7, no. 1, 2025.
- [8] S. Babaie and C. Qiao, "Towards distributed quantum error correction for distributed quantum computing," in *IEEE International Conference on Quantum Communications, Networking, and Computing (QCNC)*, 2025, pp. 66–73.
- [9] V. Guemard, "Lifts of quantum css codes," *IEEE Transactions on Information Theory*, pp. 1–1, 2025.
- [10] S. Babaie and C. Qiao, "Distributed quantum bit-phase-flip error correction based on css codes and ghz states," in *45th IEEE International Conference on Distributed Computing Systems Workshops (ICDCSW)*, 2025, pp. 195–200.
- [11] Google Quantum AI and Collaborators, "Quantum error correction below the surface code threshold," *Nature*, vol. 638, no. 8052, pp. 920–926, 2025.
- [12] J. Li, Z. Wang, A. Ferrara, Y. Gu, and R. L. Franco, "Gan decoder on a quantum toric code for noise-robust quantum teleportation," 2025. [Online]. Available: <https://arxiv.org/abs/2409.06984>
- [13] N.-H. Ji, H.-Q. Sun, B. Xiao, P.-L. Song, and H.-Y. Ma, "Quantum decoder design for subsystem surface code based on multi-head graph attention and edge weighting," *Chin. Phys. B*, vol. 34, no. 2, p. 020309, 2025.
- [14] A.-Q. Li, C.-W. Tian, X.-X. Xu, H.-Y. Ma, and J.-Q. Liang, "Global receptive field transformer decoder method on quantum surface code data and syndrome error correction," *Chin. Phys. B*, vol. 34, no. 3, p. 030306, 2025.
- [15] H. Wang *et al.*, "High-level surface code decoding via parallel ffnns on cim platforms," 2025. [Online]. Available: <https://arxiv.org/abs/2411.18090>
- [16] K. Wasa *et al.*, "Hardness of Braided Quantum Circuit Optimization in the Surface Code," *IEEE Trans. Quantum Eng.*, vol. 4, pp. 1–7, 2023.
- [17] F. Hua, Y. Chen, Y. Jin, C. Zhang, A. Hayes, Y. Zhang, and E. Z. Zhang, "Autobraid: A framework for enabling efficient surface code communication in quantum computing," in *54th Annual IEEE/ACM International Symposium on Microarchitecture*, ser. MICRO '21, New York, NY, USA, 2021, p. 925–936.
- [18] D. B. Tan, M. Y. Niu, and C. Gidney, "A SAT Scalpel for Lattice Surgery: Representation and Synthesis of Subroutines for Surface-Code Fault-Tolerant Quantum Computing," in *ACM/IEEE 51st Annual International Symposium on Computer Architecture (ISCA)*, Los Alamitos, CA, USA, 2024, pp. 325–339.
- [19] N. K. Chandra, D. Tipper, R. Nejabati, E. Kaur, and K. P. Seshadreesan, "Distributed realization of color codes for quantum error correction," *arXiv:2505.10693*, 2025.
- [20] S. de Bone, P. Möller, C. E. Bradley, T. H. Taminiau, and D. Elkouss, "Thresholds for the distributed surface code in the presence of memory decoherence," *AVS Quantum Science*, vol. 6, no. 3, 2024.
- [21] M. B. Hastings and J. Haah, "Dynamically generated logical qubits," *Quantum*, vol. 5, p. 564, 2021.
- [22] M. Kesselring, J. C. Magdalena de la Fuente, F. Thomsen, J. Eisert, S. D. Bartlett, and B. J. Brown, "Anyon condensation and the color code," *PRX Quantum*, vol. 5, no. 1, 2024.
- [23] E. Sutcliffe, B. Jonnadula, C. L. Gall, A. E. Moylett, and C. M. Westoby, "Distributed quantum error correction based on hyperbolic floquet codes," 2025.
- [24] M. S. Alam and E. Rieffel, "Dynamical logical qubits in the bacon-shor code," *arXiv:2403.03291*, 2024.
- [25] N. P. Breuckmann and J. N. Eberhardt, "Quantum low-density parity-check codes," *PRX Quantum*, vol. 2, no. 4, 2021.
- [26] S. Evra, T. Kaufman, and G. Zémor, "Decodable quantum ldpc codes beyond the $\sqrt{n}$ distance barrier using high-dimensional expanders," *SIAM Journal on Computing*, vol. 53, no. 6, pp. FOCS20–276–FOCS20–316, 2024.
- [27] T. Kaufman and R. J. Tessler, "New cosystolic expanders from tensors imply explicit quantum ldpc codes with $(n \log k n)$ distance," in *53rd Annual ACM SIGACT Symposium on Theory of Computing*, 2021, pp. 1317–1329.
- [28] M. B. Hastings, J. Haah, and R. O'Donnell, "Fiber bundle codes: breaking the $n^{1/2} \text{polylog}(n)$ barrier for quantum ldpc codes," in *Proceedings of the 53rd Annual ACM SIGACT Symposium on Theory of Computing*, ser. STOC 2021, New York, NY, USA, 2021, p. 1276–1288.
- [29] P. Panteleev and G. Kalachev, "Quantum ldpc codes with almost linear minimum distance," *IEEE Transactions on Information Theory*, vol. 68, no. 1, p. 213–229, 2022.
- [30] N. P. Breuckmann and J. N. Eberhardt, "Balanced product quantum codes," *IEEE Transactions on Information Theory*, vol. 67, no. 10, p. 6653–6674, 2021.
- [31] A. Strikis and L. Berent, "Quantum low-density parity-check codes for modular architectures," *PRX Quantum*, vol. 4, no. 2, 2023.